

CTIS 256

Final – Duration 120 Mins

Setup & Hints

- Log into “phpmyadmin”, <http://localhost/phpmyadmin>
- Username: “std” Password: “”, no password, **empty string**.
- Import “**wallet.sql**” file into “**test**” database
- There is only one table: **wallet**(code, name, opening, last, icon, amount)
 - Code: cryptocurrency codename (BTC)
 - Name: fullname (Bitcoin)
 - Opening: the price of the currency at the beginning of the day
 - Last: the current price of the currency
 - Icon: image of the currency
 - Amount: the number of currency the user owns.
- **Hints about SQL Queries:**
 - *select * from wallet*
 - *select * from wallet where code = ?*
 - *update wallet set last = ? where code = ?*
 - *update wallet set last = ?, amount = ? where code = ?*
 - *update wallet set amount = amount + ? where code = ?*
- Watch “**video.mp4**” to find out what you are supposed to do.
- You can use your own SQL commands if you want.
- You are not allowed to change structures of the database table.
- HTML Codes: ☺ : **⟳**; 📄 : **🖫**; ⚙ : **⎌**;
- Misc PHP functions :
 - **number_format(\$num, 3)** : shows 3 digits after the decimal point
 - **floatval(\$num)** : removes all leading and trailing 0s.
 - **rand(0, 10)** : generates a random integer between 0 and 10 inclusive.
- Javascript function : num.**toLocaleString()** → “1,234.45”

OBJECTIVE

You will simulate a cryptocurrency market by retrieving the user’s wallet that contains the coins of the user. It displays the user’s coins, calculates the change in percentage after the opening, and total price of the coins using the current (last) price and the amount of coins. If the user clicks on the link “Update Last Prices”, it selects at least one and at most three coins, and calculates their prices to either increase or decrease by 1% of the opening price. It calculates subtotals of each coin as well.

REQUIREMENTS

- (10Pts) The approximate page design (layout, spacing, color). Font family is 'Gill Sans'.
- (15Pts) It shows the content of the wallet in an HTML table as in Fig 1.
- (7Pts) Calculate Change (%) using last price and opening price. (green and red colors)
- (3Pts) Calculate Total column for coins.

- e. (20Pts) "Update Last Price" link selects at least one and at most three coins, and updates last prices +1% or -1% of opening price. (Example: if the opening price is 100, last price is 120, then the new price will be either 119 or 121.) (50% chance for +1% or -1%)
- f. (5Pts) It shows new price updates by a small CSS animation (green or red background color based on increase or decrease respectively)
- g. (20Pts) "Buy/Sell" column contains textboxes to add/subtract the amount of the related coin. It should use AJAX. Use 1 second delay to show spinning icon (spinner.gif). It updates amount and total columns. +3 means buy 3 coins and -3 is for selling the icon. **No need for boundary check.** Web service doesn't have to be RESTful. Use "change" event for textboxes.
- h. (20Pts) "Save Coin States" saves the content of the current table or states of the coins in a session using a stack (LIFO strategy). It basically stores the selected table states in a history. If there is at least one saved entry (coin states), "Rollback – Undo" link appears and it activates the last coin states on click. If there is nothing in the stack to undo, the button disappears. The number beside the button shows the size of the stack.



↻ Update Last Price
💾 Save Coin States

Coin			Last Price	Opening Price	Change(%)	Amount	Total	Buy/Sell
	BTC	Bitcoin	\$74,580.000	\$67,800.000	+10.00%	0.5	\$37,290.000	
	ETH	Ethereum	\$4,560.000	\$3,800.000	+20.00%	4	\$18,240.000	
	INJ	Injective	\$24.890	\$26.200	-5.00%	55	\$1,368.950	
	GRT	The Graph	\$0.288	\$0.320	-10.00%	1600	\$460.800	
	DOGE	Dogecoin	\$0.184	\$0.160	+15.00%	5000	\$920.000	
	RNDR	Render Token	\$14.420	\$10.300	+40.00%	320	\$4,614.400	
	SOL	Solana	\$143.310	\$168.600	-15.00%	60	\$8,598.600	

Fig1: Page Initially

Attention: Partial grade is applicable only for running codes.

- Good Luck -